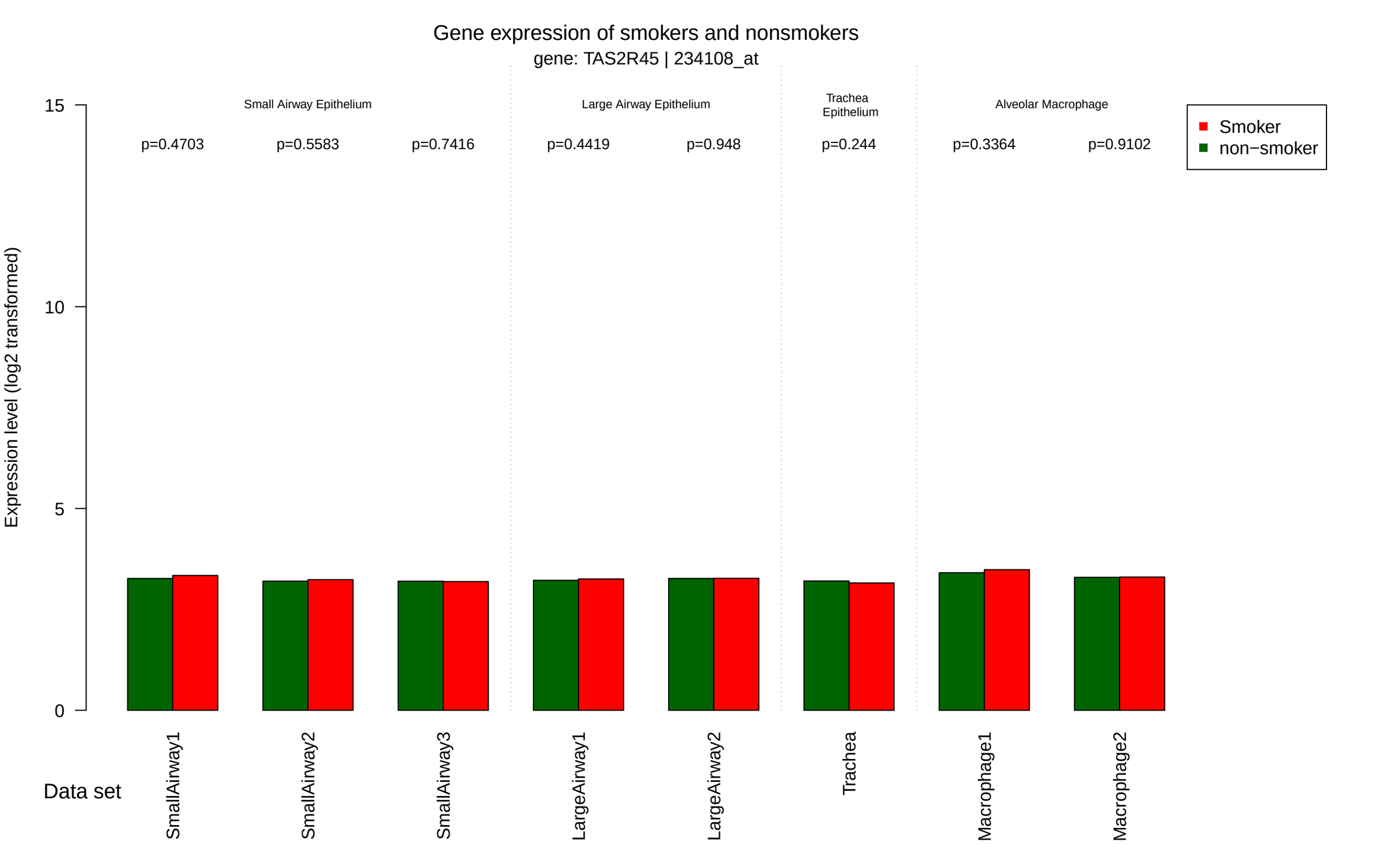
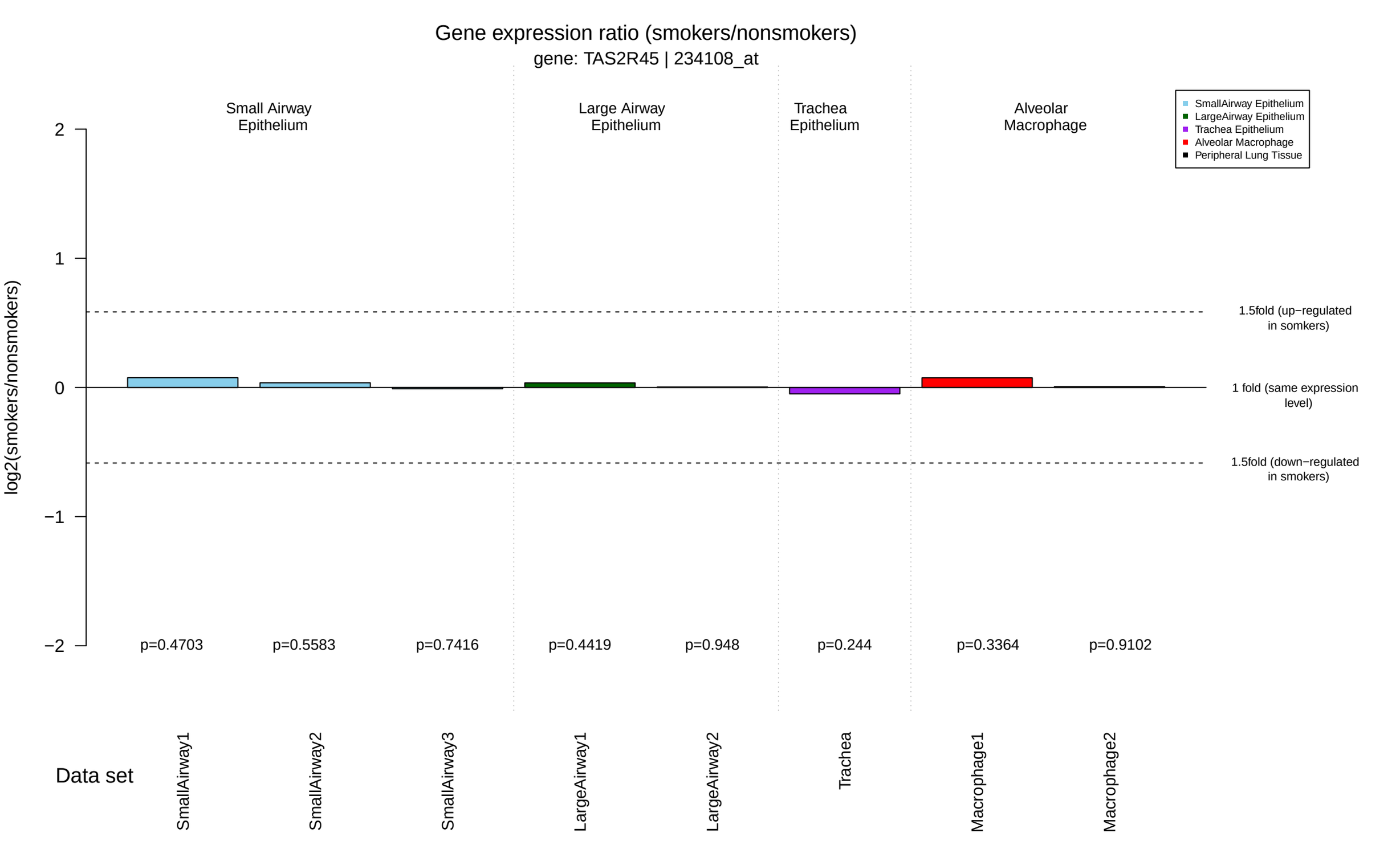
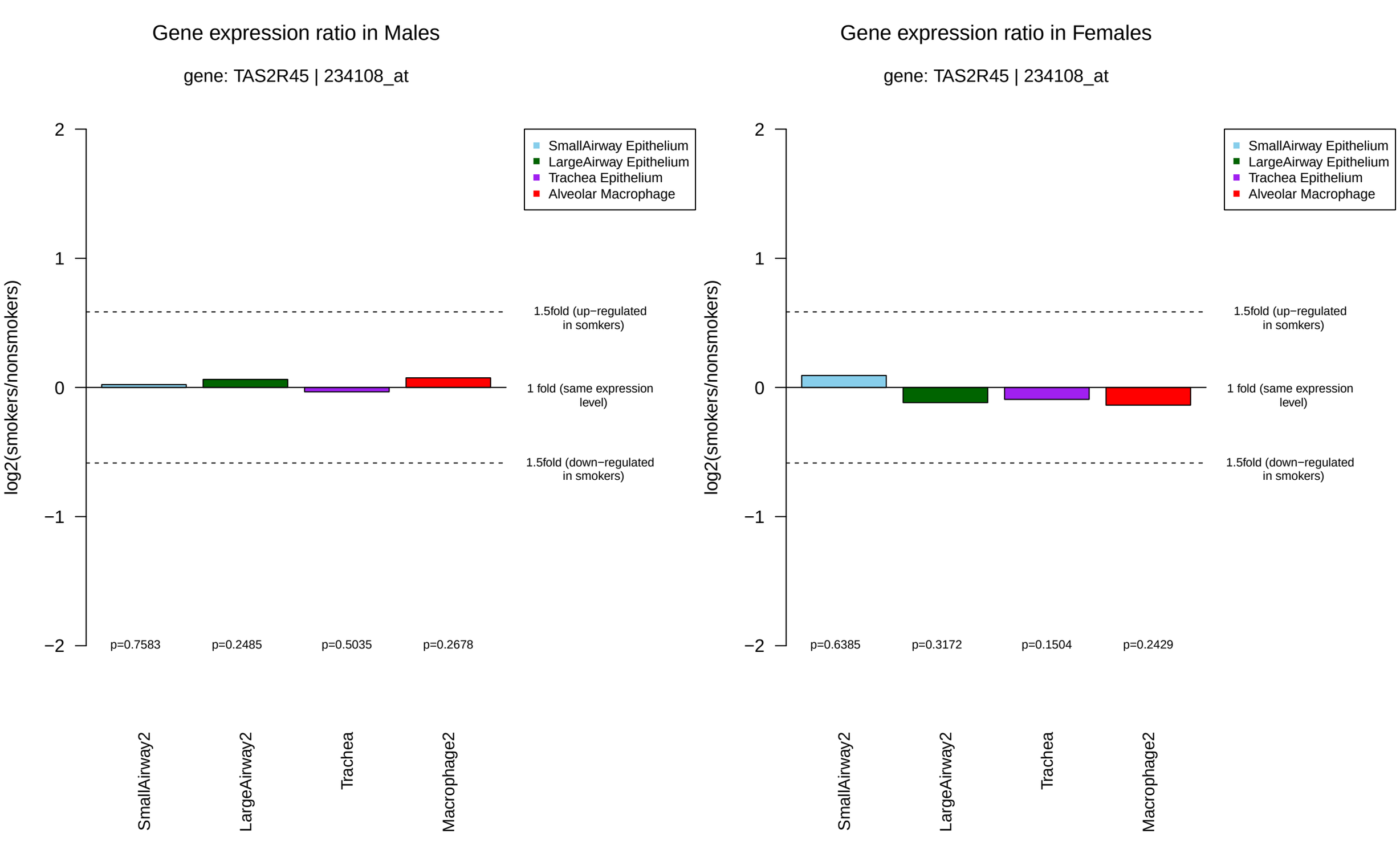
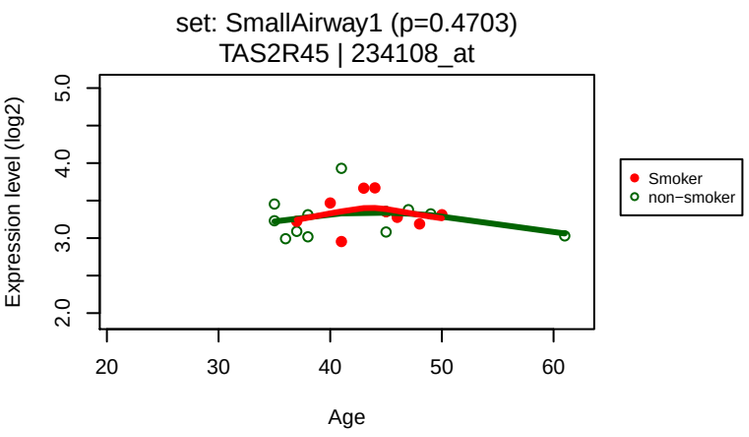
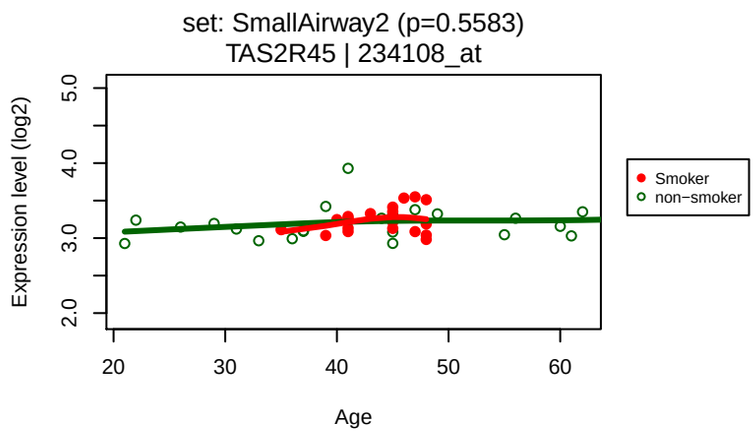
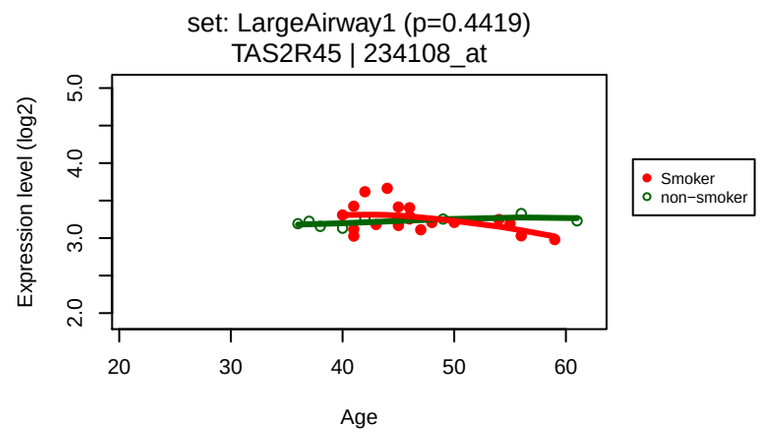

Gene expression of smokers and nonsmokers
gene: TAS2R45 | 234108_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4703
p=0.5583
p=0.7416
p=0.4419
p=0.948
p=0.244
p=0.3364
p=0.9102
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

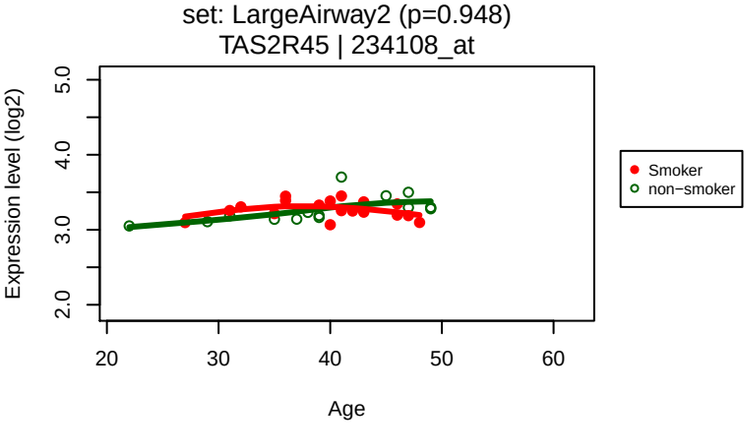

Gene expression ratio (smokers/nonsmokers)
gene: TAS2R45 | 234108_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4703
p=0.5583
p=0.7416
p=0.4419
p=0.948
p=0.244
p=0.3364
p=0.9102
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

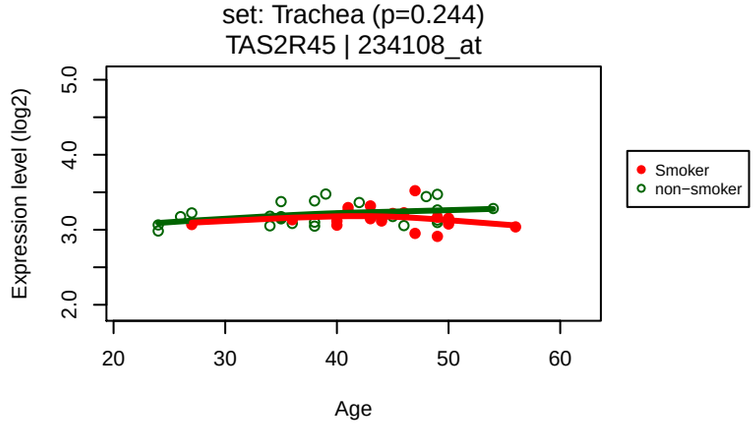

Gene expression ratio in Males
gene: TAS2R45 | 234108_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7583
p=0.2485
p=0.5035
p=0.2678
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TAS2R45 | 234108_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6385
p=0.3172
p=0.1504
p=0.2429
SmallAirway2
LargeAirway2
Trachea
Macrophage2

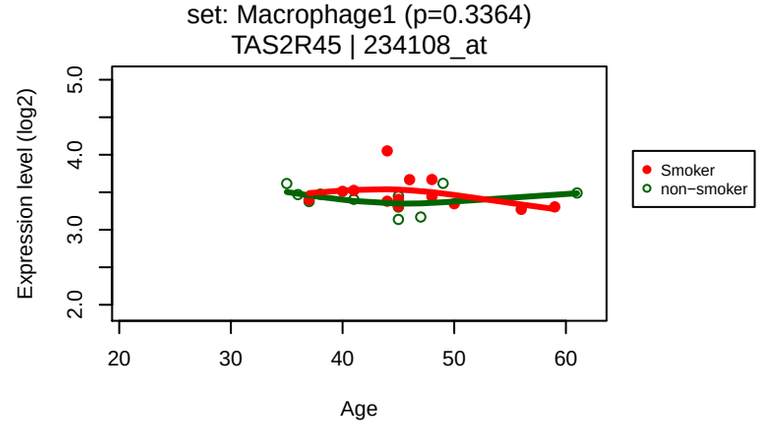

set: SmallAirway1 (p=0.4703)
TAS2R45 | 234108_at
Expression level (log2)
Age
Smoker
non−smoker

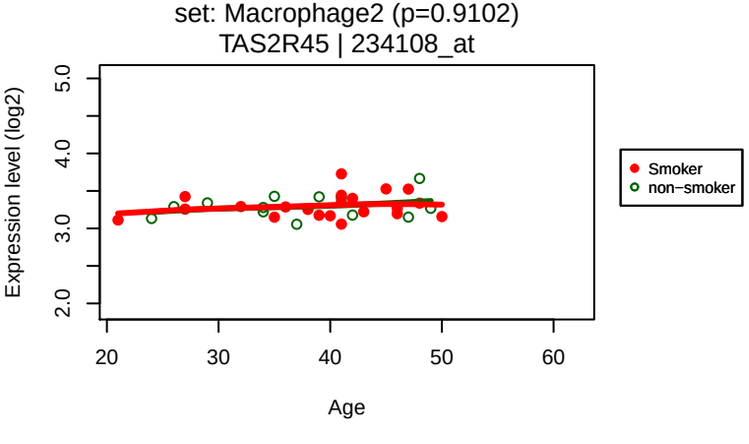

set: SmallAirway2 (p=0.5583)
TAS2R45 | 234108_at
Expression level (log2)
Age
Smoker
non−smoker

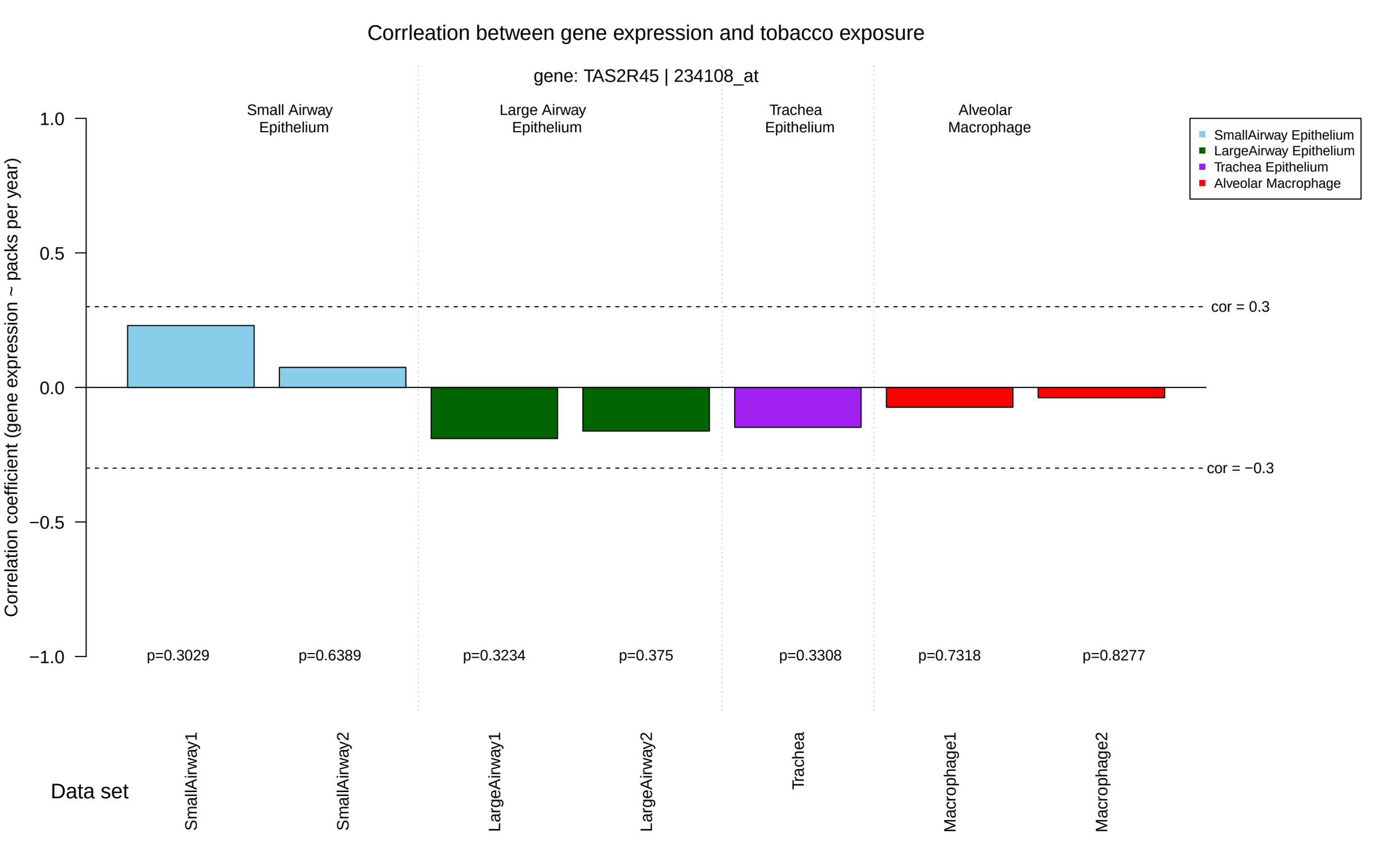

set: LargeAirway1 (p=0.4419)
TAS2R45 | 234108_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.948)
TAS2R45 | 234108_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.244)
TAS2R45 | 234108_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3364)
TAS2R45 | 234108_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9102)
TAS2R45 | 234108_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TAS2R45 | 234108_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3029
p=0.6389
p=0.3234
p=0.375
p=0.3308
p=0.7318
p=0.8277
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2